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Effects of MgO on dielectric properties and electrical conductivity of ternary zinc magnesium phosphate glasses

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ABSTRACT

Glasses with composition $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ (x = 5, 8, 13, 18 and 20 mol%) have been successfully prepared by the melt-quenching technique. The dielectric permittivity (ε') and loss factor (ε'') were measured in the frequency range of 0.01 Hz to 1 MHz and in the temperature range of 303 to 573 K. From the results, there is evidence of dipolar relaxation occurring between 10^3–10^6 Hz, while at low frequencies, the spectrum is dominated by dc conduction which was manifested by the $1/\omega$ slope of the loss factor plot. The value of the relaxing frequency (ω_p) plotted against $1/T$ shows a single relaxation mechanism with an activation energy of 0.45 eV. The average value of the activation energy for dc conduction was much higher (1.25 eV) suggesting its diffusion movement had encountered more difficult steps than the small displacement changing dipoles. With increasing MgO concentration, the dielectric permittivity (ε'), dc conductivity (σ_{dc}) and dielectric strength ($\Delta\varepsilon$) decrease and these were attributed to some of the magnesium ions participated in the glass-forming positions as well as modifiers. At lower temperatures, the complex permittivity plots present a skewed arc with center point lying below the real axis which is a non-Debye characteristic. The empirical data were sufficiently fitted by using the Harviliak–Negami equation. The temperature dependent of the parameter α is discussed.

1. Introduction

Phosphate glasses have several advantages over conventional silicate and borate glasses due to their superior physical properties such as high thermal expansion coefficient, low melting and softening temperatures and high ultra-violet and far infrared transmissions [1]. Among oxide glasses, phosphate glasses are both scientifically and technologically important materials because of their structural versatility to accept several cation and/or anion exchanges. These features allow the reengineering of glass formulation, which leads to advances in their physical properties and optimizing of processing parameters [2]. In many technology applications, glass must serve as an electrical insulator or conductor. Mott and Davis [3] discussed abundantly about the properties of non-crystalline materials with the movement of electrons as its origin. However, for the study, the glass systems is considered as ionic glasses due to the existence of alkaline earth ions (Mg^{2+}) and transition metal ions (Zn^{2+}) with no more than one valence state. Hence, electrons jumping from low valency state to other with higher valence state are not taking place. Glass is a complex material and until today there has been no theory of ionic transport which is accepted widely. Thus, ionic diffusion in ionic conducting glass material has been an issue of interest because of its importance in technological uses. Therefore, a better understanding of its electrical conductivity is required [4].

Phosphate glass structure is strongly dependent on the ratio of its oxygen and phosphate (O/P ratio) as set by glass composition. In the present study, the O/P ratio of the glass is at a constant value of ~2.88 and the structure is mainly composed of ultraphosphate, Q^3 and Q^2 units.

The study on dielectric properties such as dielectric constant, loss and ac conductivity of phosphate glasses over a wide range of frequency and temperature is expected not only to reveal comprehensive information regarding the nature and origin of the loss occurring in these materials as well as conduction mechanism but also provide information on the structural aspect of the glasses [5,6]. The dielectric properties of usual interest are the real (ε') and imaginary (ε'') components of the complex permittivity $\varepsilon^*(\omega) = \varepsilon'(\omega) - i\varepsilon''(\omega)$. The permittivity of a material reflects the molecular relaxation and transport processes of the material which depends on many parameters such as temperature, time and pressure [7].

The analysis of measured complex permittivity is usually performed by a curve-fitting technique based on, typically, one of the following models: Debye, Cole–Cole, Davidson–Cole,

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Havriliak–Negami (HN) and Kolhrausch–Williams–Watts (KWW). If several relaxation regions are observed in the available frequency window, the HN-function can be used to describe as well as to separate the different relaxation processes [8]. Moreover, HN-function is the most conventional empirical and analytical dielectric expression to describe the generalized broadened and asymmetric relaxation loss peak observed in many dielectric materials over a wide frequency range [9].

Kordes et al. [10] classified $BeO–P_2O_5$, $MgO–P_2O_5$ and $ZnO–P_2O_5$ glasses as anomalous glass. They exhibited unusual behavior in the evolution of their physical properties (mass density, refraction index) at the metaphosphate composition [11]. Not many literature can be found discussing the dielectric properties in detail for the binary $MgO–P_2O_5$ [12,13] and $ZnO–P_2O_5$ glasses [14]. In addition, it is of interest to investigate the physical, refractive and dielectric properties of the simultaneous admixture of MgO and ZnO into the phosphate compositions. To investigate the transition of these properties into the ultraphosphate composition, a series of ternary glasses with O/P < 3 has been synthesized. In addition, other properties such as complex permittivity and electrical conduction have also been examined. The present study reports the dielectric properties and conductivity performed on $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses over a wide range of compositions, temperatures and frequencies.

2. Materials and methods

The glasses were prepared by melt-quenching technique using chemical powders, phosphorus (V) oxide (99.99% purity), magnesium oxide (98% purity) and zinc oxide (99.7% purity). Ternary $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ was synthesized for $5 \leqslant x \leqslant 20$ (x gives the content of MgO in mole basis). Stoichiometric mixture of ZnO, MgO and P_2O_5 were weighed and mixed thoroughly in a 50 cm^3 alumina crucible in appropriate quantities to constitute a 15–20 g batch. The crucible was covered and heated in an electric furnace for about one hour at a temperature of 673 K to allow the phosphate to decompose and react with other batch constituent before melting occur, ordinarily. The crucible was then transferred to another electric furnace and kept at 1473 K for 2 h. The melt was stirred occasionally every 20 min to ensure homogeneity and proper mixing. Next, each melt was cast by pouring it into a preheated stainless steel cylindrical two-split mould to form glass rods about 20 mm long and 10 mm in diameter. After casting, each glass was immediately transferred to an annealing furnace, and held at temperature of 673 K for 1 h and slowly cooled to room temperature. The amorphous nature of these glass samples was established and confirmed using X-ray diffraction (XRD). The glasses that were successfully prepared were transparent and free of bubbles. The nominal composition of glasses in this study is listed in Table 1 with the designation for the respective mole fraction.

Compositional analysis of the glasses was performed by Laser Induce Breakdown Spectroscopy (LIBS). The disk of each glass was striked with laser at five different positions. Analysis revealed the existence of the expected elements that constitute the glass samples.

Density (ρ) of the prepared glass samples was measured at room temperature by standard Archimedes method (apparent weight loss) using acetone as the immersion fluid. The sample was first weighed in air, g_a, and then in an immersion liquid (acetone) g_L, with density, $\rho_L = 0.789$ g cm^{-3}. The weighing process was measured by using electronic balance. The density of the sample was then calculated using the following relationship

$$\rho = \frac{g_a \rho_L}{(g_a - g_L)}. \tag{1}$$

The estimated error was ±0.001 g cm^{-3}. The molar volume (V_m) of each sample was calculated from the relation $V_m = M_T/\rho$. M_T is the total molecular weight of the glass system taken as

$$M_T = X_{ZnO}Z_{ZnO} + X_{MgO}Z_{MgO} + X_{P_2O_5}Z_{P_2O_5}, \tag{2}$$

where X_{ZnO}, X_{MgO} and $X_{P_2O_5}$ are the mole fraction of the constituent oxide and Z_{ZnO}, Z_{MgO} and $Z_{P_2O_5}$ are the molecular weight of the constituent oxide.

Samples for electrical property measurements were prepared by cutting from annealed glass bars into disk-form and both faces of the samples were finely polished using 1500-grit, 2400-grit and 4000-grit polishing paper in order to get ≈0.5 mm thickness with smooth and parallel surfaces. The thickness of the glass specimens was measured using a digital micrometer gauge. Pure aluminum (99.999%) was evaporated onto both sides of the samples as electrodes, 10 mm in diameter by using a thermal evaporator (Edwards Auto 306). These samples were stored in a desiccator and taken out only at the time of measurement of their properties.

The study of dielectric properties were made using a High-Resolution Dielectric Analyzer (Novocontrol) connected with a BDS 1200 sample holder over a frequency range of 10^{-2} Hz to 10^6 Hz with an applied potential of 1 V and in the temperature range from 303 to 573 K. The temperature was controlled to ±0.1 K by Novotherm controller unit. A software package WinDeta control, calculate and record all measurements.

The experimental data was fitted using Havriliak–Negami dielectric relaxation function by means of statistic software (WinFit) for non-linear curve fitting. Mean square deviation (MSD) was obtained from the statistical analysis which is the mean value of the quadratic deviations in y-direction of each data point with the fit function.

3. Results

3.1. Dielectric studies

The empirical data were sufficiently fitted by using Havriliak–Negami (HN) dielectric relaxation function, superimposed by a conductivity term as shown below:

$$\varepsilon^* = \varepsilon' - i\varepsilon'' = \sum_{k=1}^{m} \left\{ \frac{\Delta\varepsilon_k}{[1 + (i\omega\tau_k)^{\alpha_k}]^{\beta_k}} + \varepsilon_{\infty_k} \right\} - i\left(\frac{\sigma_\circ}{\varepsilon_\circ \omega}\right)^n, \tag{3}$$

Table 1
Composition and selected properties of the $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses.

Sample code	Composition (mol%)			Molar ratio	Density, ρ (g cm^{-3})	Molar volume, V_m (cm^3 mol^{-1})	E_ω (eV)	E_σ (eV)
	ZnO	MgO	P_2O_5	O/P	(±0.001 g cm^3)	(±0.001 cm^3 mol^{-1})	(±0.03)	(±0.05)
CZ 05	30	5	65	2.77	3.228	36.768	0.40	1.11
CZ 08	30	8	62	2.80	3.118	37.008	0.51	1.08
CZ 13	30	13	57	2.88	2.937	37.643	0.51	1.13
CZ 18	30	18	52	2.96	2.815	37.469	0.41	1.04
CZ 20	30	20	50	3.00	2.712	38.143	0.41	1.16

where $\sigma_\circ$ is the dc conductivity, $\Delta\varepsilon_k$ is the dielectric strength of the kth relaxation, τ_k is the relaxation time of the kth component, α is the width parameter, β is the asymmetry parameter, ε_{∞_k} is the infinite permittivity of the kth relaxation and n is the exponential factor. The HN equation is reduced to the Davidson–Cole expression when $\alpha = 1$ and to Cole–Cole function when $\beta = 1$. The Debye equation is obtained if both α and β are equal to unity.

Fig. 1(a) and (b) shows the frequency dependence of the real (ε') and imaginary (ε'') parts of the complex permittivity for CZ 13 measured at different temperatures. The variation of response towards frequencies and temperatures are almost similar for all glass samples in the present study. Fig. 2(a) shows the result of real (ε') as a function of temperature for the sample $(ZnO)_{30}(MgO)_{13}(P_2O_5)_{57}$ at different frequencies. Fig. 2(b) represents real (ε') as a function of MgO concentration measured at 1 kHz and 473 K.

3.2. Electrical conductivity studies

The values of the relaxing frequency (ω_p) and dc conduction (σ_{dc}) for different composition of $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses plotted as a function of reciprocal temperature, 1000/T is presented in Fig. 3(a) and (b). Table 1 presents the values of activation energy for relaxation process (E_ω) and dc conduction (E_σ) determined from those slopes.

Fig. 4(a) shows the dependence of dc conductivity, σ_{dc} at difference temperature as a function of MgO content for all glasses. Fig. 4(b) presents the plot of dielectric strength, $\Delta\varepsilon$ as a function of MgO concentration at 303 K.

Fig. 5(a) shows the complex permittivity plot for 18 mol% MgO measured from 303 to 423 K. With increasing temperature the spike rises up due to the increase in dc conduction, as shown in in-

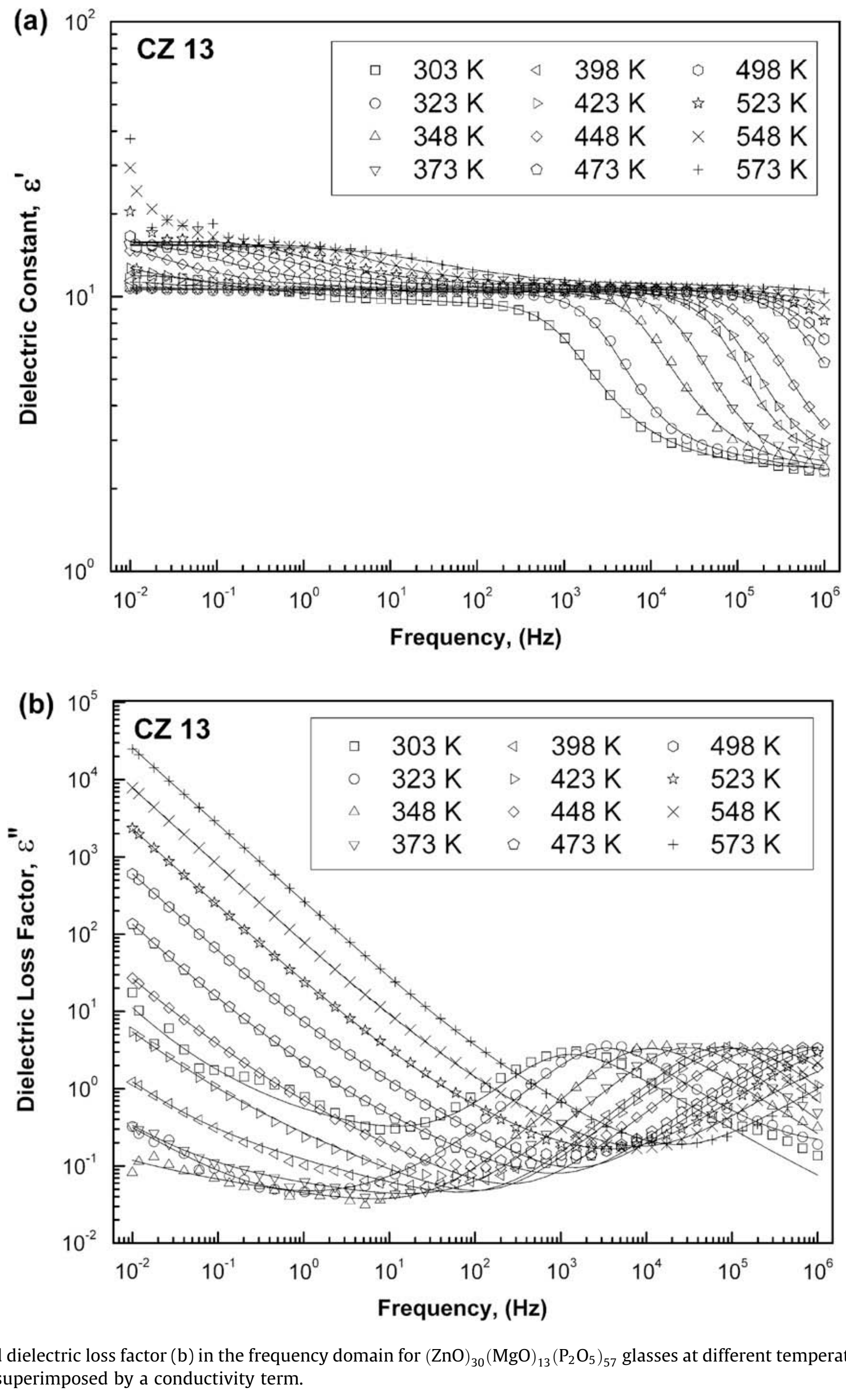

Fig. 1. Dielectric constant (a) and dielectric loss factor (b) in the frequency domain for $(ZnO)_{30}(MgO)_{13}(P_2O_5)_{57}$ glasses at different temperatures. Solid line indicates fitting by the Havriliak–Negami equation superimposed by a conductivity term.

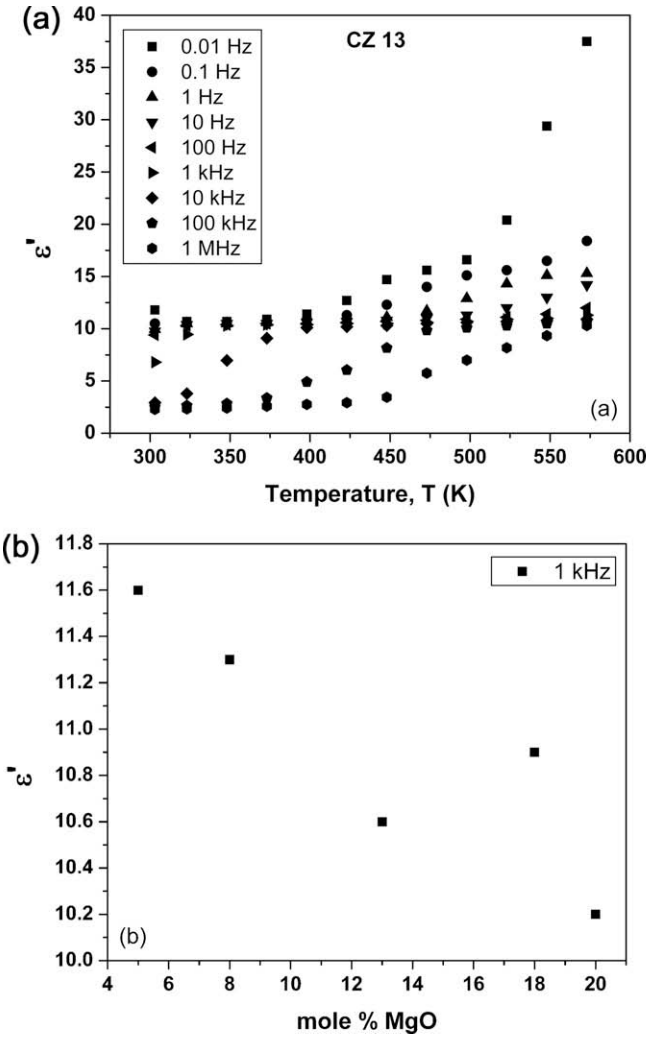

Fig. 2. (a) Temperature dependence of the dielectric constant, ε' at different frequencies for $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$. (b) Composition dependence of the dielectric constant measured at 1 kHz at temperature 473 K.

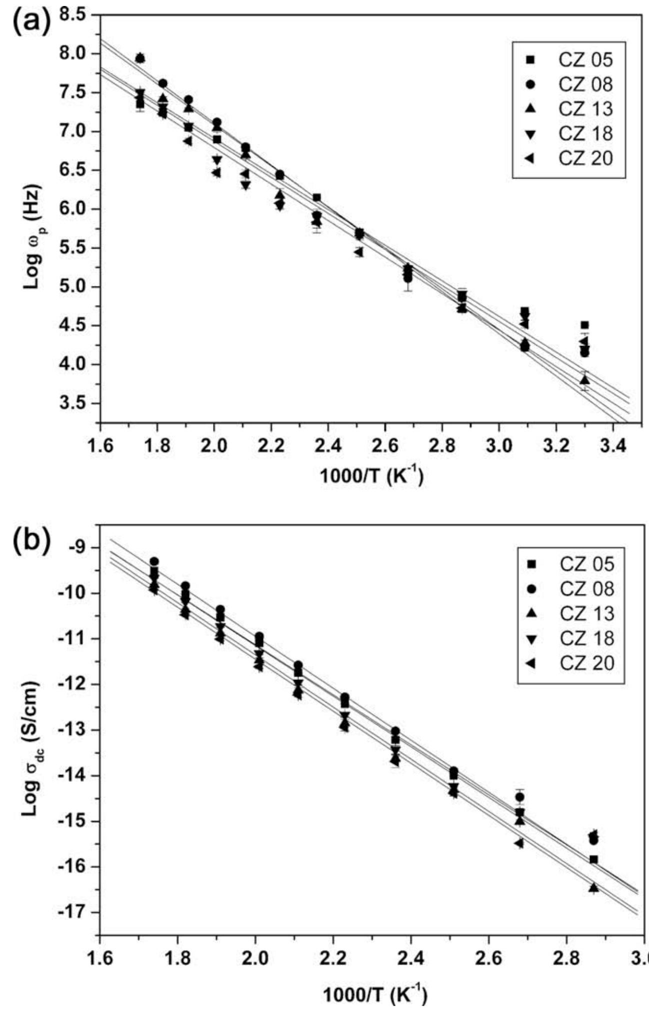

Fig. 3. (a) Temperature dependence of relaxation frequency and (b) dc conductivity for $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses with different MgO concentrations.

set in Fig. 5(a). Fig. 5(b) is a representative complex permittivity plot for the CZ 05 glass at 423 K which display clearly that the empirical data was sufficiently fitted by using Eq. (3) and their corresponding electrical equivalent circuits is shown as well.

The variation of alpha (α) values with the different composition of $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses as a function of temperature is presented in Fig. 6.

4. Discussion

4.1. Dielectric studies

In a dielectric study, the real part of permittivity (ε') represents the polarizability of the material, while the imaginary part (ε'') represents the energy loss due to polarization and ionic conduction [15]. For ionic conducting glasses its dielectric properties is mainly due to ionic motions.

It is well known that charge carriers in glass cannot move freely through a glass matrix but they can be displaced and polarized as a response to an applied alternating field. Owen [16] pointed out that dipoles or relatively immobile ions which also take part in network-forming may also contribute to the electrical properties. The significance of the complete dipolar response can be interpreted in the region near the loss peak frequency, ω_p which is dominated by the dipolar transitions [17]. As observed in ε' spectra from 10^{-2} to 10^6 Hz shown in Fig. 1(a), there exists two relaxation dispersions, one in the low frequency region (<1 kHz) and the other in the high frequency region (>1 kHz). The latter is the main relaxation dispersion which originates from the bulk polarization. For frequency, $\omega < \omega_p$, ε' is a plateau indicating that the immobile and mobile ions are able to oscillate and respond with the applied field and thus contribute fully to the total polarization. At higher frequency limit ($\omega > \omega_p$), where the immobile and mobile ions cannot oscillate as quickly as the applied field, dispersion takes place and ε' decreases to a constant value, ε'_∞ which probably results from a rapid polarization process [18]. As observed, the step-like decrease shifts uniformly in the range of 10^3–10^6 Hz as temperature increases indicating a thermally-activated process. At lower frequencies (<1 kHz) and temperatures above 448 K, ε' approaches a limiting low frequency plateau which we associate with space charge polarization and dc conduction, respectively. However, the strong increase of ε' at even lower frequencies and in temperatures excess of 498 K may be attributed to localized motion of mobile ions in glassy network. Since the glass matrix is random, the high value of ε' in the low frequency limit is attributed to the charge carriers diffused along barriers of varying heights, but tends to pile up at high free energy barriers in the field direction. This leads to a net polarization of the ionic medium, contributing to the high value of ε' at low frequencies [19].

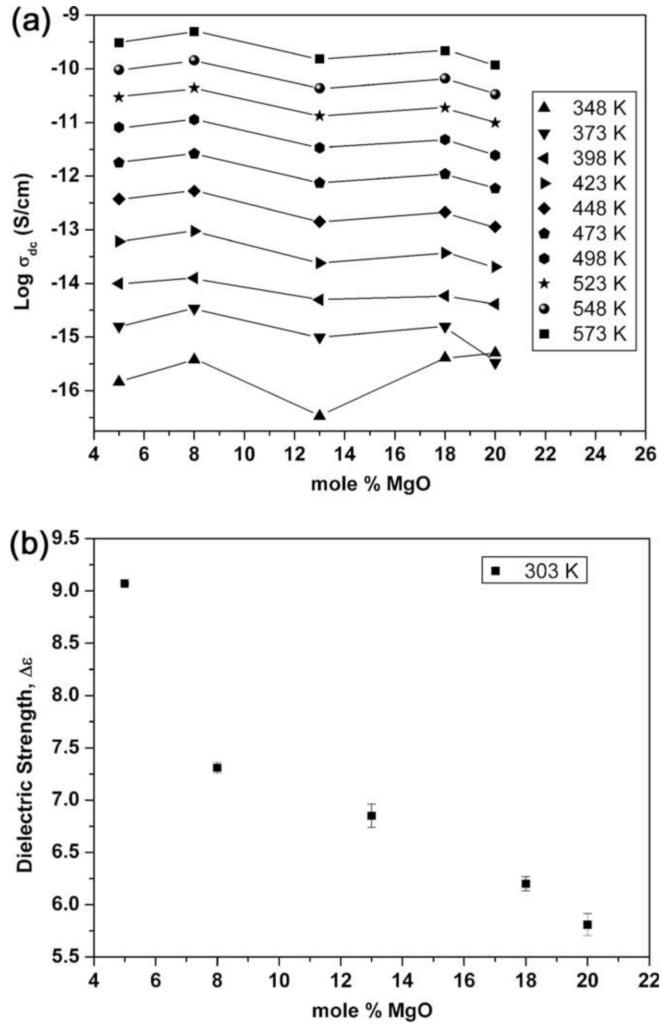

Fig. 4. (a) The variation of dc conduction (solid lines are the guides to the eye) and (b) the dielectric strength, $\Delta\varepsilon$, as a function of MgO concentration for $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses at different temperatures and room temperature, respectively.

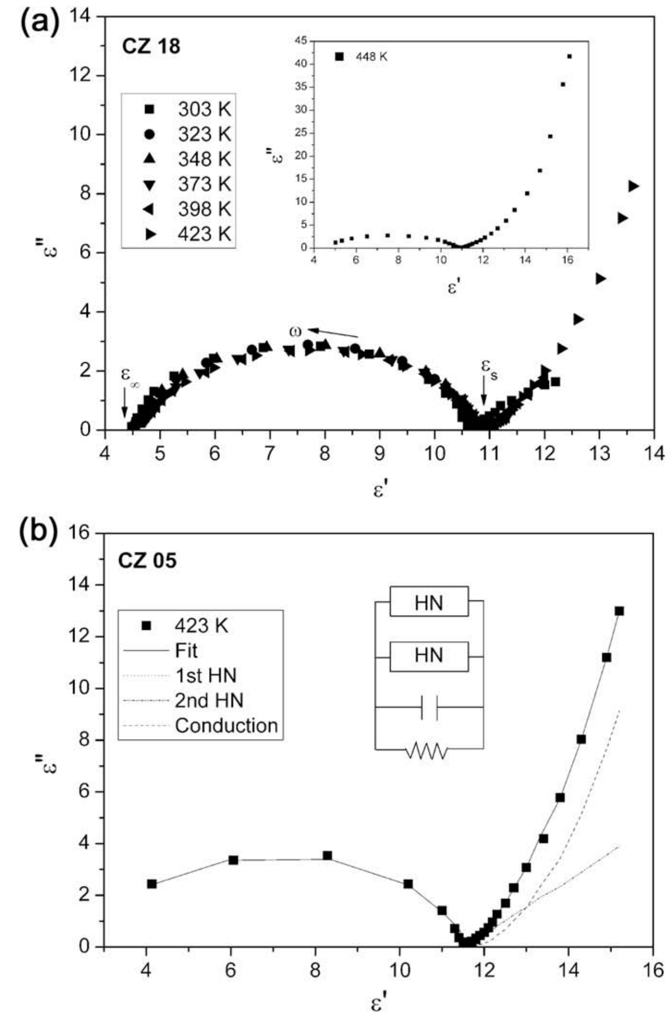

Fig. 5. Complex permittivity plots for $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses with (a) $x = 18$ at different temperatures. Inset represents the complex permittivity plot at higher temperature. (b) Two HN-functions and a conductivity contribution were fitted to the whole spectrum. Corresponding equivalent circuit is shown.

At low frequencies, the ε'' spectrum is dominated by dc conduction which is clearly evidenced by the $1/\omega$ slope of the loss factor plot with frequencies as shown in Fig. 1(b). ε'' is found to increase with increasing temperature and dielectric loss peak at higher frequency range has been observed which is an evidence of dipolar type relaxation. Generally, the dielectric relaxation effects are observed only when the metal ions present in the glass network are in the divalent state [20,21]. Among the constituent's elements in these glasses, the divalent zinc ions having higher electronegativity may form dipoles with a pair of cationic vacancies and such types of dipole are responsible for the dielectric relaxation effects [22]. The position of these loss peaks gradually shifted toward higher frequencies with increasing temperatures indicating that the time for ions to reach a stable state is reduced, i.e., the mean relaxation rate has increased. However, the magnitude of the maximum loss is unaffected by the measurement temperatures. For the corresponding loss peak, the dependence of ε'' on frequency is associated with damping effect as dipolar polarization occurs. This means that some energy is absorbed from the field [23,24]. Above the peak of ε'' the decrease in high frequency regime is attributed to high ionic inertia, impeding the dipole to follow the frequency response, i.e., energy loss decreases. However, the loss peak which corresponds to the additional relaxation process at relative low frequencies region cannot be seen in the measured ε'' data on account of the dominant influence of the conduction losses.

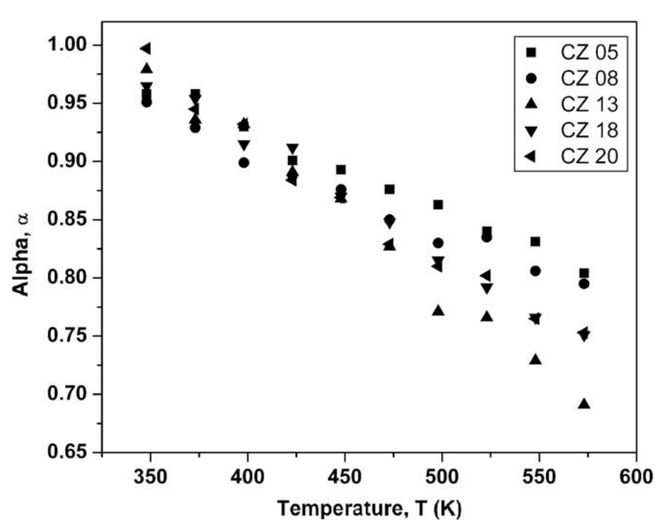

Fig. 6. The plots of alpha versus temperature for different MgO concentration.

From Fig. 2(a), it is clear that ε' increases with increase in temperature, the increase being different at different frequencies. All the studied samples showed the same behavior. In general, the spectrum increases slightly with increasing temperature which is usually associated with decreases in bond energies. In other words, the glassy network relaxes and weakens the intermolecular forces as temperature increases and hence the ions displacement i.e., orientation polarization becomes easier [25,26]. From the spectra, two relaxation processes can be observed. In region I, the relaxation process is observed at all the measured temperatures range as a step. The step shifts to a higher temperature as frequency increases from 10 Hz onwards. The relaxation process corresponds to a dipolar-like relaxation process. Similarly, the second relaxation process (in region II) also appears as a step from 448 K to 573 K as the frequency varies from 0.01 to 10 Hz. However, the corresponding dielectric loss maxima cannot be seen. In region III, the value of ε' becomes larger at lower frequencies and at higher temperatures is not an indication for spontaneous polarization [26] but is attributed to electrode polarization effect that is associated to the accumulation of space charges in sample-electrode interface.

From Fig. 2(b), the parameter ε' is shown in a decreasing trend which shows Mg^{2+} ions to occupy preferentially substitutional positions in the glass network for higher concentrations of MgO [27,28]. This reduces the polarization as well. The participation of Mg^{2+} in the glass-forming position is also pointed out by Suzuya et al. [29]. For low concentrations of MgO, the Mg^{2+} ions tend to occupy the interstitial positions of the glass and contribute to the polarization [28].

4.2. Electrical conductivity studies

The ε'' as a function of frequency shown in Fig. 1(b) displays maximum at a characteristic frequency at each measured temperature. The characteristic frequency also corresponds to the relaxation frequency, ω_p and can be extracted by means of a curve-fitting analysis. The value of the ω_p shifted towards a higher frequency when the temperature increased and can be expressed by the usual Arrhenius relation: $\omega_p = \omega_\circ \exp(-E_\omega/kT)$, where E_ω is the activation energy for relaxation process, k the Boltzmann constant and $\omega_\circ$ the pre-exponential factor. From Fig. 3(a), values of the relaxing frequency (ω_p) plotted against $1000/T$ show a linear line indicating only one electrical transportation mechanism exists. The increasing trend of ω_p with temperature is reasonable because the driving force for ion displacement is heat, and the kinetic energy of ions becomes larger, resulting in faster movement as temperature increases. Consequently, relaxation time, τ decreases and shift the relaxation frequency, ω_p to a higher frequency [30]. From Fig. 3(b) the dependence of dc conduction on $1000/T$ for different composition of $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses show that the conductivity of the studied glasses is also a thermally-activated process obeying the Arrhenius relation: $\sigma = \sigma_\circ \exp(-E_\sigma/kT)$, where E_σ is activation energy of dc conduction. The values of E_ω and E_σ for different concentrations of MgO have been calculated from the slopes of Fig. 3(a) and (b), respectively and tabulated in Table 1. Error for calculating the activation energy was calculated using least-square fit. The values of E_σ are much larger than E_ω indicating that both the conduction and relaxation processes are due to the different conduction mechanisms and carriers. In this case, the most probable carrier for dc conduction is the relative free mobile ions, Mg^{2+}, while the most probable for displacement curve is contributed by the Zn^{2+} ions. This is expected because of the mobile ions that contributed to dc conduction by mean of diffusion movement encountered more difficult steps than the small displacement changing dipoles [31].

The activation energy, E_ω, for different MgO concentrations is tabulated in Table 1. The value of E_ω in the glass network does not show any dependencies with MgO concentration and varies between 0.40 to 0.51 eV. The activation energy, E_σ is also summarized in Table 1 for the different glass compositions. Although the values of E_σ varies dramatically, there is certainly an increasing trend as the concentration of MgO increases. The variations of the activation energies and dc conductivity in a non-linear behavior was pointed by Prasad and Radhakrishna [32] to be the results of the existence of inhomogeneous ionic clusters due to the structural effects produced by different ratios of modifier to former.

Fig. 4(a) shows that σ_{dc} decreases as the MgO content is raised. $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ glasses have a complex composition and are an admixture of network formers, intermediates and modifiers. In the present system with the amount of ZnO fixed, the higher the MgO concentration, the lower is the P_2O_5 concentration. Normally, NBO site increases with increasing modifier content. Stanworth [33] and Sun [34] pointed out that zinc ions with higher electronegativity (1.6 for Zn, 1.2 for Mg) and bond strength (3.12 eV for Zn, 1.61 eV for Mg) belong to the intermediate class of glass-forming oxide. Moreover, the calculated fractional ionic character value (FIC) 0.57 and 0.84 for the chemical bonds of Zn–O and Mg–O, respectively indicates that Zn^{2+} cation are able to occupy network-forming sites as well as interstitial positions in glasses, while, cation Mg belongs to the category of ions which can occupy interstitial positions in glasses [13]. However, Suzuya et al. [29] work suggests that Mg^{2+} cation acts as a network former as well. Therefore, with further increase of ZnO or MgO content, these atoms tend to participate in the glass network and the structure expands [35]. This expectation is in line with the results shown in Table 1 which shows decreases in density and increases in molar volume as the content of MgO increases.

As the MgO content increases, some of the Mg^{2+} cation may take part in the glass-forming position as well as Zn^{2+} cation. Consequently, the amount of relatively free mobile ions (Mg^{2+}) is reduced and σ_{dc} decreases. Meanwhile, the residual Zn^{2+} located in the interstices of the considered glass matrix impedes the conduction pathway. Moreover, the temperature is also a contributing factor to σ_{dc}. The rise of σ_{dc} is attributed to the long-range migration of mobile ions in the glass matrix by means of thermally-activated hopping movements. As the thermal energy in the material is gradually enhanced, more and more charge carriers are successful in overcoming the binding energy i.e., attraction force among the glass networks and able to diffuse to longer distances [36].

The dielectric strength values defined by $\Delta\varepsilon = \varepsilon_s - \varepsilon_\infty$ can be extracted from the fitting of the Havriliak–Negami equation. The dielectric strength of glass represents the maximum polarizability that it can withstand in the applied electrical field without breaking down. This is important when the glass is used as a substrate for electrical or electronic devices. Polarizability is greatest at those frequencies at which all dipoles can oscillate in response to the field. From Fig. 4(b) the decrease of $\Delta\varepsilon$ with the increase of MgO concentration indicates that dipolar density contribution to the polarization has been reduced. This is expected since the amount of Mg^{2+} cations residing at the interstitial positions in the glass is reduced, therefore, the concentration of the dipoles formed by the Mg cations and the non-bridging oxygens is decreased [28].

In Fig. 5(a) at lower temperatures the complex permittivity plots for 18 mol% MgO show a skewed arc with center point lying below the real axis by an angle $\alpha\pi/2$ which is a non-Debye characteristic. This behavior is a clear evidence of existing distribution of relaxation time and may have resulted from the wide range of forces that are required to restrain the orientation motion of dipoles [37]. In addition, a spike was observed at the onset of low frequency as temperature increased. The spike is a clear evidence of conduction which was overlapping with electrode polarization. Fig. 5(b) is a representative plot of the glasses which displays clearly that the empirical data was sufficiently fitted by using Eq. (3) and adequately modeled by assuming the equivalent circuit with two HN terms, a capacitance and resistance which is in parallel combination. The capacitance, C and resistance, R, described the high frequency permittivity and dc conduction, respectively. The main arc which is represented by the first HN-function is equal to the main loss peak in the dielectric loss factor plot. This arc which is gradually suppressed as temperature increases is associated with the conduction loss. This behavior is due to the conduction loss becoming the main contributor and appears to be much greater than the relaxation loss. The second HN-function is associated to electrode polarization. The arc intersecting ε' axis at lower value represents ε_∞, while the one that intersects at higher value represents ε_s. The difference $\varepsilon_s - \varepsilon_\infty$ (or relaxation intensity) increases with temperature irrespective of MgO concentration. This behavior may attribute to the increase in free charges resulting from thermal effects, which were blocked in greater numbers at the interfaces, thus increasing the aptitude of the dipoles to be polarized [38].

The values of α and β from Eq. (3) are extracted from the curve-fitting analysis of the experimental data for ε'' spectra. The values of α and β parameter range between $0 \leqslant \alpha < 1$ and $0 < \beta \leqslant 1$. As observed, the value of alpha decreases with increasing temperatures. The parameters α and β are related to the limiting behavior of the complex dielectric function at low and high frequencies: $\varepsilon_s - \varepsilon'(\omega) \sim \omega^{\alpha}$; $\varepsilon'' \sim \omega^{\alpha}$ for $\omega \ll \omega_p$ and $\varepsilon'(\omega) - \varepsilon_\infty \sim \omega^{-\alpha\beta}$; $\varepsilon'' \sim \omega^{-\alpha\beta}$ for $\omega \gg \omega_p$, with ω_p being the angular frequency at the ε'' maximum [22]. In other words, parameters α and β provide the width and the asymmetric, respectively, of the ε'' peak in a log–log plot for the HN-function. In the present case, as shown in Fig. 6 the decrease of α parameter with increasing temperature indicates the broadening of relaxation peaks. This represents an increase in the damping effect [39]. Generally, the value of α increases with decreasing internal degrees of freedom of the dipoles, i.e., high viscosity and therefore interactions between dipoles become weak. In the present case, as the temperature increases the value of α decreases, irrespective of MgO concentration. This is because as the temperature is getting higher, the viscosity is getting lower. Likewise, the dipoles obtain more energy to vibrate, i.e., internal degree of freedom of the dipoles increases. Consequently, interaction between dipoles increases [30,37].

5. Conclusions

The dielectric permittivity and loss factor of $(ZnO)_{30}(MgO)_x(P_2O_5)_{70-x}$ with different MgO concentrations are frequency and temperature dependent. From the thorough analysis of the experimental data, dc conduction loss dominates at low frequencies and dipolar type relaxation occurs at high frequency regions as observed in the dielectric loss factor plot. Dielectric permittivity, dielectric strength and dc conductivity decreases with an increase in MgO concentration. Such a behavior is attributed to the dual role played by magnesium ions that participated in the glass-forming positions in addition to modifier. The dielectric responses of all glasses studied were able to be represented by two Havriliak–Negami relaxation processes superimposed with a conductivity term. From the spectra curve-fitting, values of the parameters $\sigma_\circ$, α and β were extracted.

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